

Miss Dua Agha**Discrete Structures - CT-162****Question 1****[CLO 1 - 15 marks]**

- a. Translate the following statement into propositional logic and construct a truth table for it:
 "If the alarm is triggered, then either the security guard responds or the police are called. However, if the alarm is triggered and the security system is not working, then the police are called if and only if the security guard does not respond."
(7 Marks)

Using proposition: A: Alarm is triggered, G: Security guard responds, P: Police are called, S: Security system is working.

- b. Illustrate that $(P \rightarrow Q) \wedge (\neg P \rightarrow R) \equiv (P \wedge \neg R) \rightarrow Q$ are logically equivalent by developing a series of logical equivalences.
(8 Marks)

Question 2**[CLO 2 - 15 marks]**

- a. Let S be the set of all people. Identify which of the following relations on S are equivalence relations, which are partial order relations, or neither. For each relation, justify your answer by identifying which properties of an equivalence relation or partial order relation they fail to satisfy.

1. $R_1 = \{(x, y) \mid x \text{ and } y \text{ have the same birth year}\}$
2. $R_2 = \{(x, y) \mid x \text{ has lived in every city that } y \text{ has lived in}\}$
3. $R_3 = \{(x, y) \mid x \text{ and } y \text{ have exactly the same zodiac sign}\}$
4. $R_4 = \{(x, y) \mid x \text{ and } y \text{ have at least one common city of residence}\}$ **(7 Marks)**

- b. Let $A = \{a, b, c, d\}$. A relation R on A is defined as

$$R = \{(a, a), (b, b), (d, d), (a, b), (b, a), (a, c), (c, a), (c, b), (b, c)\}$$

Draw the directed graph (digraph) of R and hence write down its corresponding matrix representation with respect to the ordering (a, b, c, d) , and determine which of the properties are satisfied by R: reflexive, symmetric, antisymmetric, and transitive.
(8 Marks)

Question 3**[CLO 3 - 15 marks]**

- a. Prove by mathematical induction that

$$3 + 3 \cdot 5 + 3 \cdot 5^2 + \dots + 3 \cdot 5^n = \frac{3(5^{n+1} - 1)}{4}$$

for all integers $n \geq 0$. **(7 Marks)**

- b. Analyze the recursive sequence f defined by $f(0) = f(1) = 3$ and for $n = 0, 1, 2, \dots$, use the given recurrence relation to find $f(2), f(3), f(4)$, and $f(5)$.

$$1. f(n+1) = 3f(n) + 7$$

$$2. f(n+1) = \frac{f(n)}{f(n-1)}$$

(8 Marks)**Question 4****[CLO 3 - 15 marks]**

- a. A university club has 8 science students and 6 arts students. A committee of 5 students is to be formed, and different roles will be assigned within the committee. **(9 Marks)**

Compare the following:

1. How many ways can a 5-member committee be formed if it must contain exactly 3 science students and 2 arts students?
2. From all 14 students, a president, vice-president, secretary, treasurer, and coordinator are to be selected. How many ways can these 5 distinct positions be assigned?
3. How many ways can a 5-member committee be formed from all 14 students, and then a president and secretary be chosen from within the committee?

- b. What is the probability that a five-card poker hand contains a straight, that is, five cards that have consecutive kinds? (Note that an ace can be considered either the lowest card in an A-2-3-4-5 straight or the highest card of a 10-J-Q-K-A straight.) **(6 Marks)**